Spatial Demographic Change in Parker County,Texas (2014 - 2024)

A Tract-Level Analysis of Income, Population Growth, and Demographic Shifts

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Summary

This project examines demographic and economic change at the census tract level in Parker County, Texas between 2014 and 2024 using American Community Survey (ACS) data.

The analysis focuses on identifying spatial patterns in income growth, population expansion, and demographic change to better understand how the county is evolving over time.

Key Findings

- Median household income increased across most tracts, with widening disparities between high- and low-income areas.
- Population growth accelerated after 2019 and concentrated in eastern and southeastern areas.
- The Hispanic population share increased across much of the county.
- The population is gradually aging, with growth in the 65+ population.
- Spatial inequality is increasing alongside suburban expansion.

Together, these findings indicate that Parker County is undergoing rapid suburban transformation characterized by economic growth, demographic diversification, and increasing spatial stratification.

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1. Introduction & Purpose

Parker County, Texas has experienced rapid growth over the past decade, accompanied by visible changes in housing development, infrastructure, and community composition. While population increases and new developments are frequently highlighted, these narratives often lack a detailed, tract-level perspective on where and how changes are occurring.

This project analyzes demographic and economic change at the census tract level between 2014 and 2024 using American Community Survey (ACS) data.

The goal is to provide a data-driven understanding on how Parker County is evolving by examining:

- Median household income growth, identifying where incomes are rising or stagnating
- Population growth and distribution, highlighting areas of concentrated expansion
- Racial and ethnic shifts
- Age structure trends, including patterns of aging and youth population changes

By combining spatial visualization with tract-level analysis, this report highlights both the areas of rapid growth and emerging inequalities shaping the county. Overall, the analysis reveals patterns of economic upgrading, demographic diversification, and suburban expansion.

2. Data Sources

- American Community Survey (ACS) 5-Year Estimates (2014, 2019, 2024)
- U.S. Census TIGER/Line Shapefiles

Variables Used:

- Median household income
- Total population
- Age distribution
- Racial/ethnic composition
- School enrollment

3. Tools & Methods

Tools & Packages

- R / RStudio
- tidyverse
- tidycensus
- sf
- tigris
- ggplot2
- ggspatial
- scales

Methods

1. Data Collection
    - Retrieved tract-level ACS data via API for 2014, 2019, and 2024
    - Ensured reproducibility through scripted queries
2. Data Cleaning & Validation
    - Harmonized variables across years
    - Addressed missing values and inconsistencies
    - Verified tract-level structure and geometry
3. Exploratory Data Analysis (EDA)
    - Generated summary statistics
    - Examined distributions & trends
    - Created line plots and boxplots
4. Spatial Analysis & Visualization
    - Produced tract-level maps of income, population, and demographic change
    - Added contextual layers (roads, cities, county boundaries)
    - Created side-by-side comparisons across years

4. Results

Income Growth

Median household income increased substantially across nearly all census tracts between 2014 and 2024. The overall distribution shifted upward, with the most significant gains occurring between 2019 and 2024. By 2024, several tracts experienced increases exceeding $40,000-$60,000, while the gap between the highest- and lowest-income tracts widened considerably.

Income growth was not uniform. Higher-growth eastern and southeastern tracts experienced the largest gains, forming a clear spatial pattern of economic concentration.

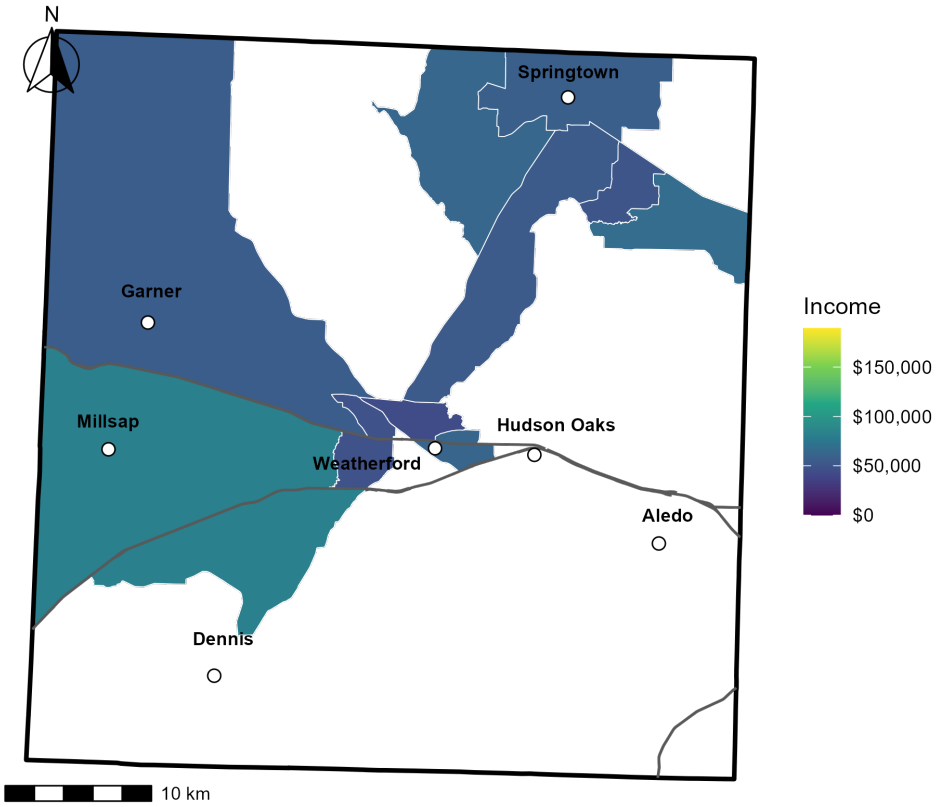
Most tracts experienced positive income growth, though the magnitude varies widely. A small number of high-growth tracts account for a disproportionate share of total gains, highlighting emerging income disparities.

These patterns suggest:

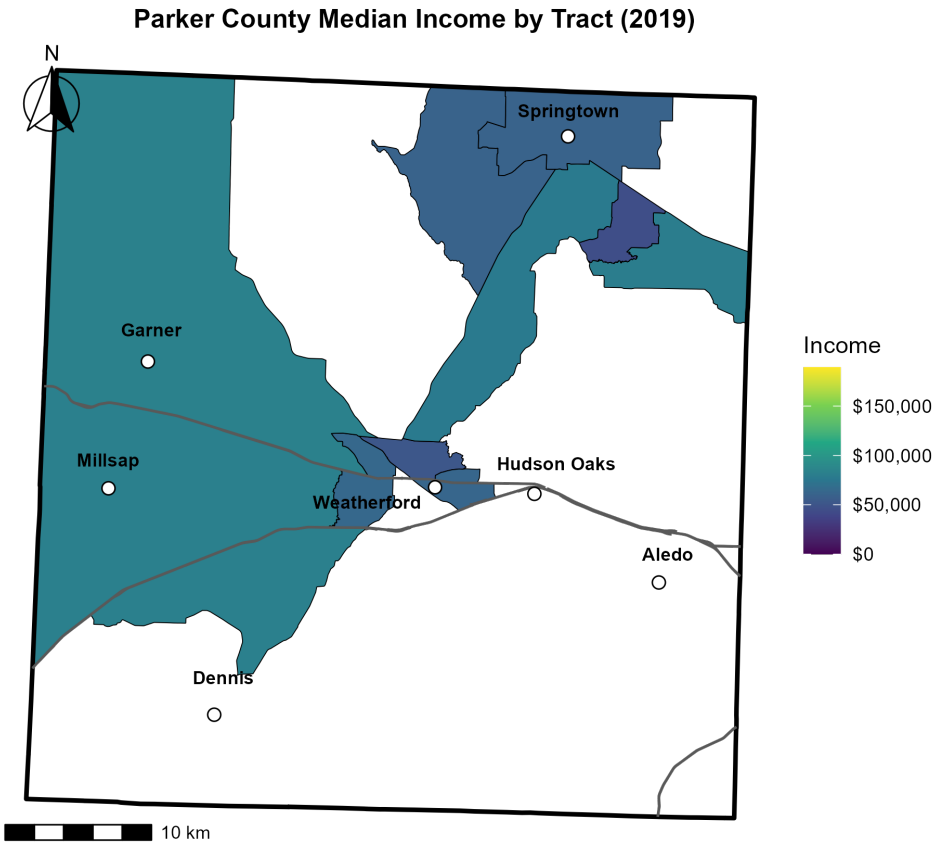
- Rapid economic upgrading in high-growth suburban corridors.
- Increasing spatial inequality across the county.
- Emerging economic stratification between regions.

If these current trends continue, Parker County may experience rising housing costs with rising property tax pressures, reduced affordability in high-growth areas and greater separation between rural western and affluent eastern tracts.

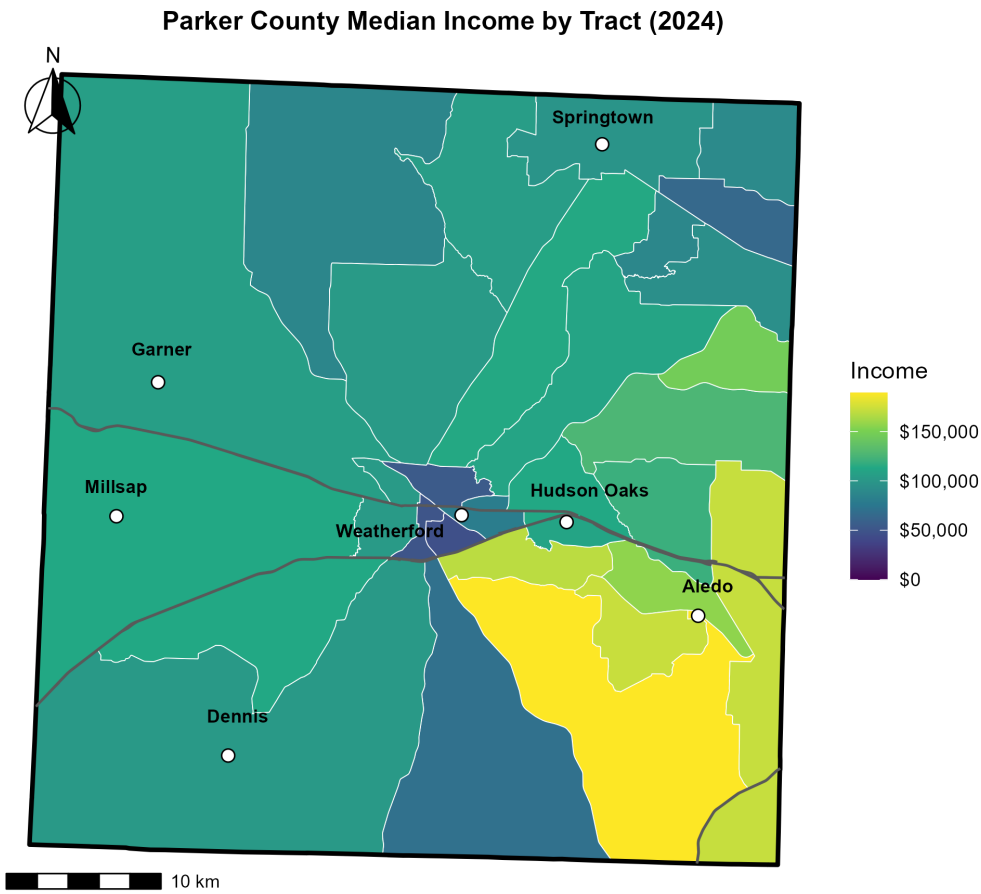
Parker County Median Income by Tract (2014)



Note: White areas represent tracts created after 2014 under 2024 tract boundaries and therefore have no income estimates for that year.
Source: American Community Survey (ACS) 5-Year Estimates



Note: White areas represent tracts created after 2019 under 2024 tract boundaries and therefore have no income estimates for that year.
Source: American Community Survey (ACS) 5-Year Estimates



Source: American Community Survey (ACS) 5-Year Estimates

```
knitr::include_graphics(c("../outputs/med_income_plot.png",
                          "../outputs/income_change_plot.png"))
```

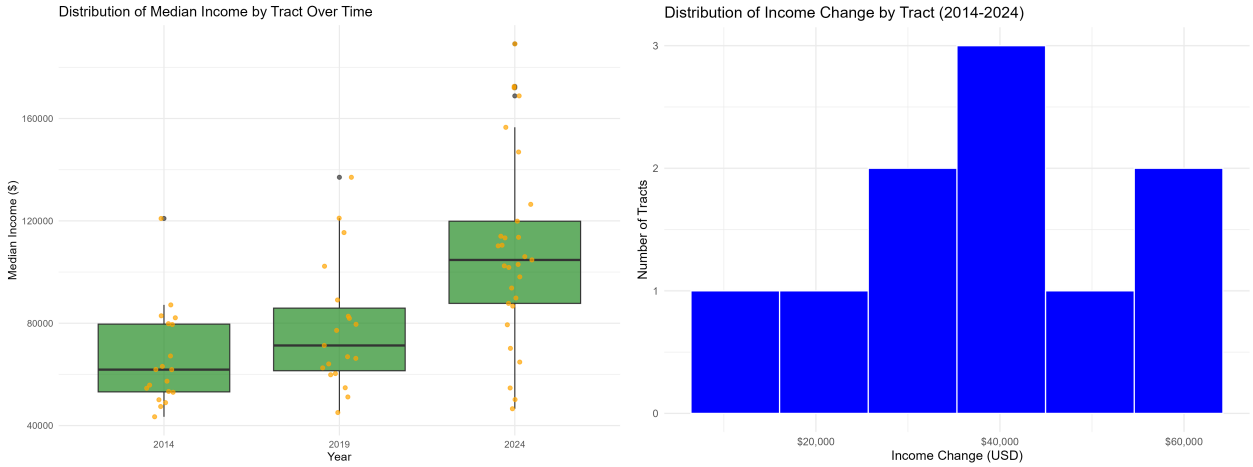


Figure 1: Income Distribution Patterns

Population Growth

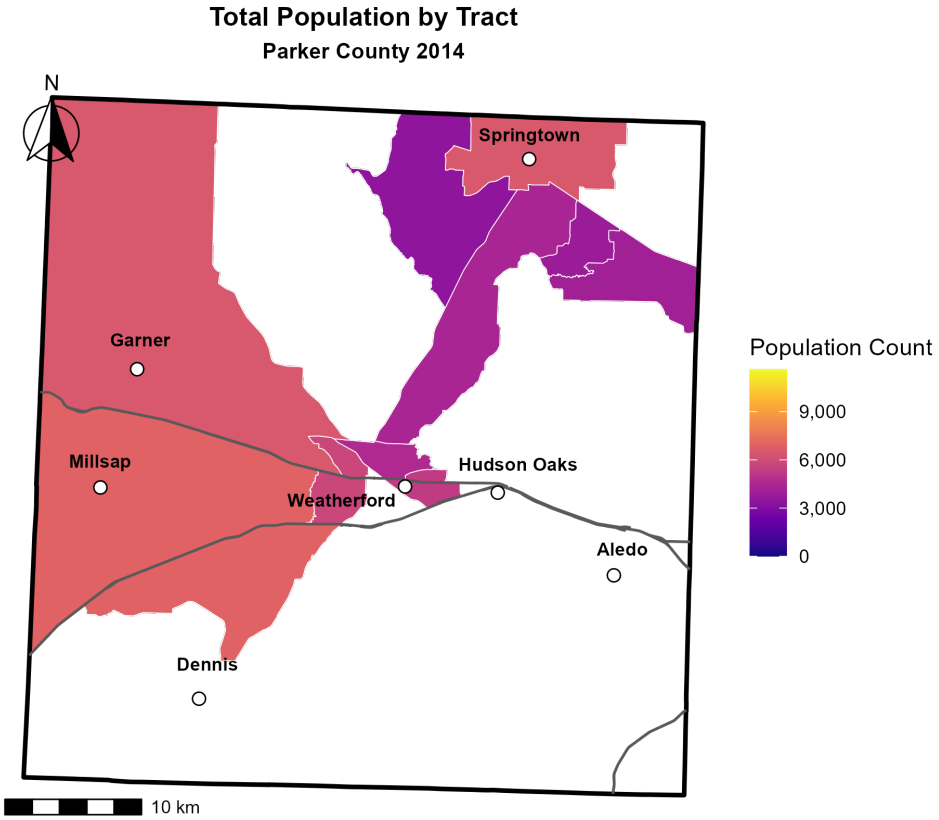
Total population increased steadily from 2014 to 2019 and accelerated significantly between 2019 and 2024.

Growth is concentrated in tracts near the edge of the Dallas-Fort Worth metropolitan and significant growth is noted in some western tracts, reflecting outward suburban expansion.

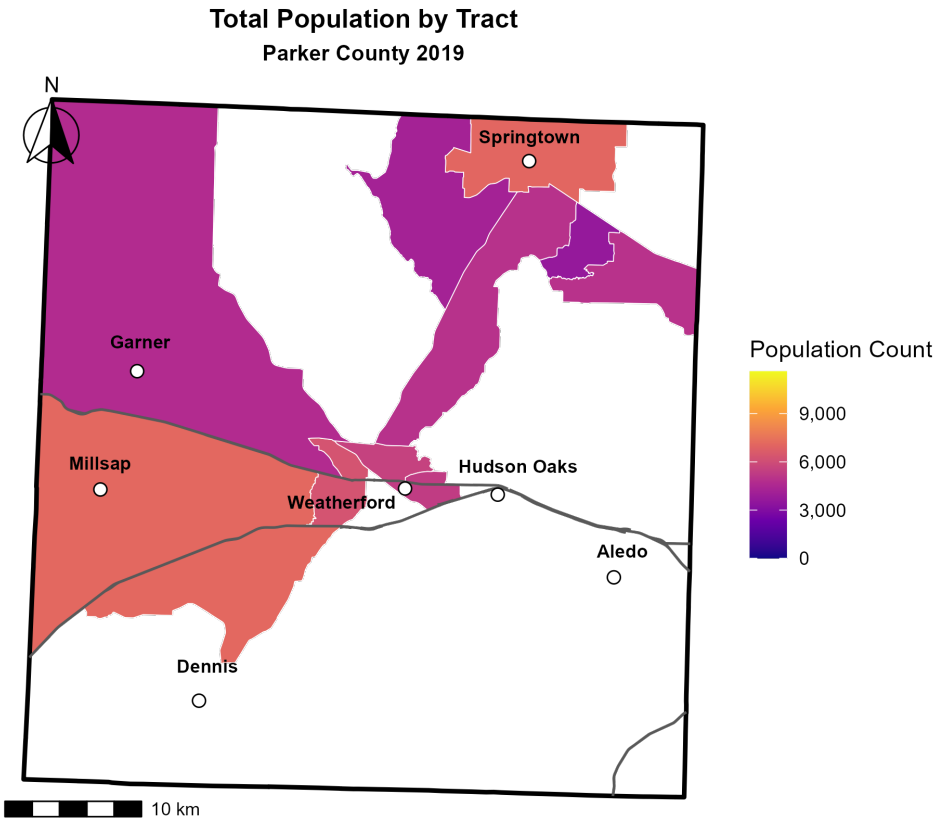
The creation of new census tracts during this period highlights the scale of growth and indicates structural changes in the county's geographic organization.

These results suggest:

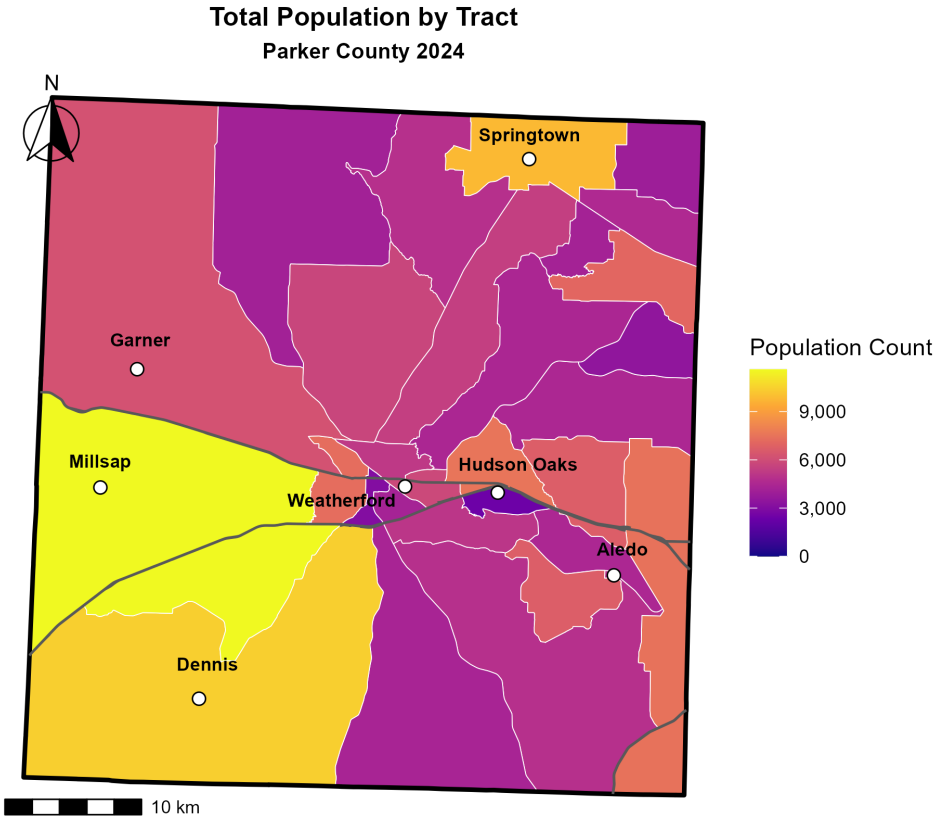
- Growth is reshaping the county's spatial structure
- Expansion is driven by metropolitan spillover
- Infrastructure and housing demand are increasing rapidly.



Note: White areas represent tracts created after 2019 under 2024 tract boundaries and therefore have no population estimates for that year.
Source: American Community Survey (ACS) 5-Year Estimates



Note: White areas represent tracts created after 2019 under 2024 tract boundaries and therefore have no population estimates for that year.
Source: American Community Survey (ACS) 5-Year Estimates



Note: White areas represent tracts created after 2019 under 2024 tract boundaries and therefore have no population estimates for that year.
Source: American Community Survey (ACS) 5-Year Estimates

Demographic Shifts

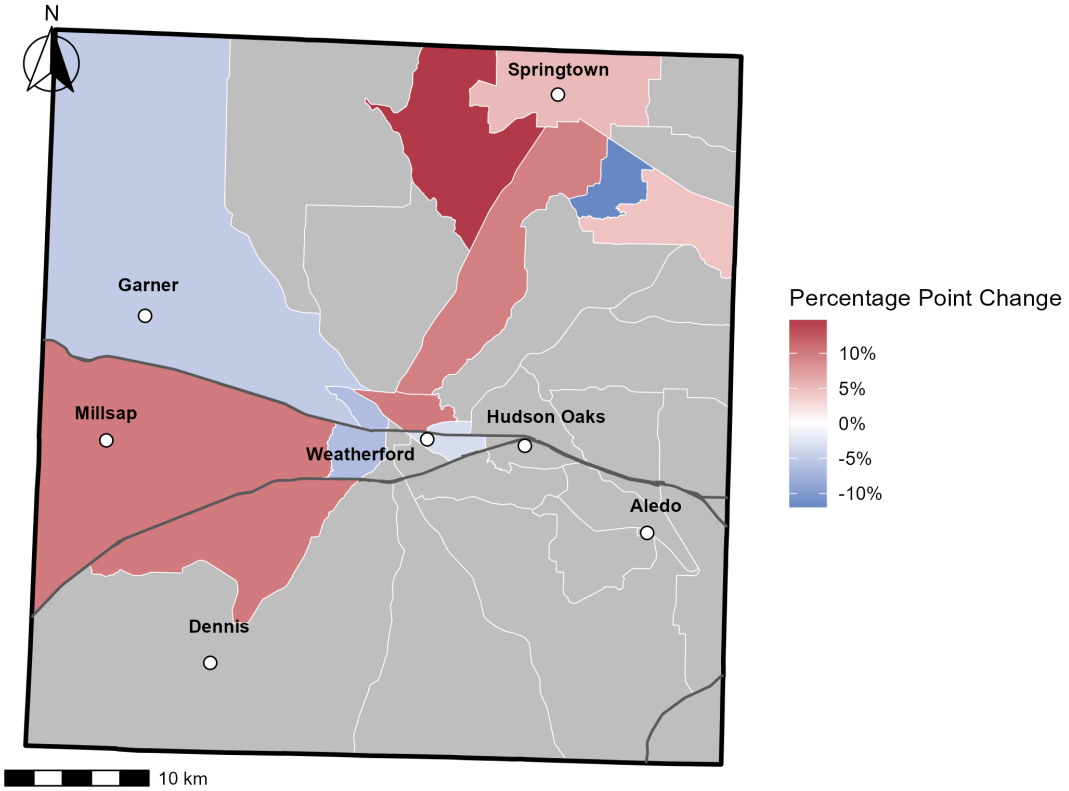
The Hispanic population share increased across most tracts between 2014 and 2024, while the White population share declined proportionally. Asian and Black population shares showed modest but consistent increases. Variation across tracts also widened, indicating increasing demographic differentiation.

These results suggest that Parker County is becoming more demographically diverse, reflecting broader suburban diversification trends across Texas.

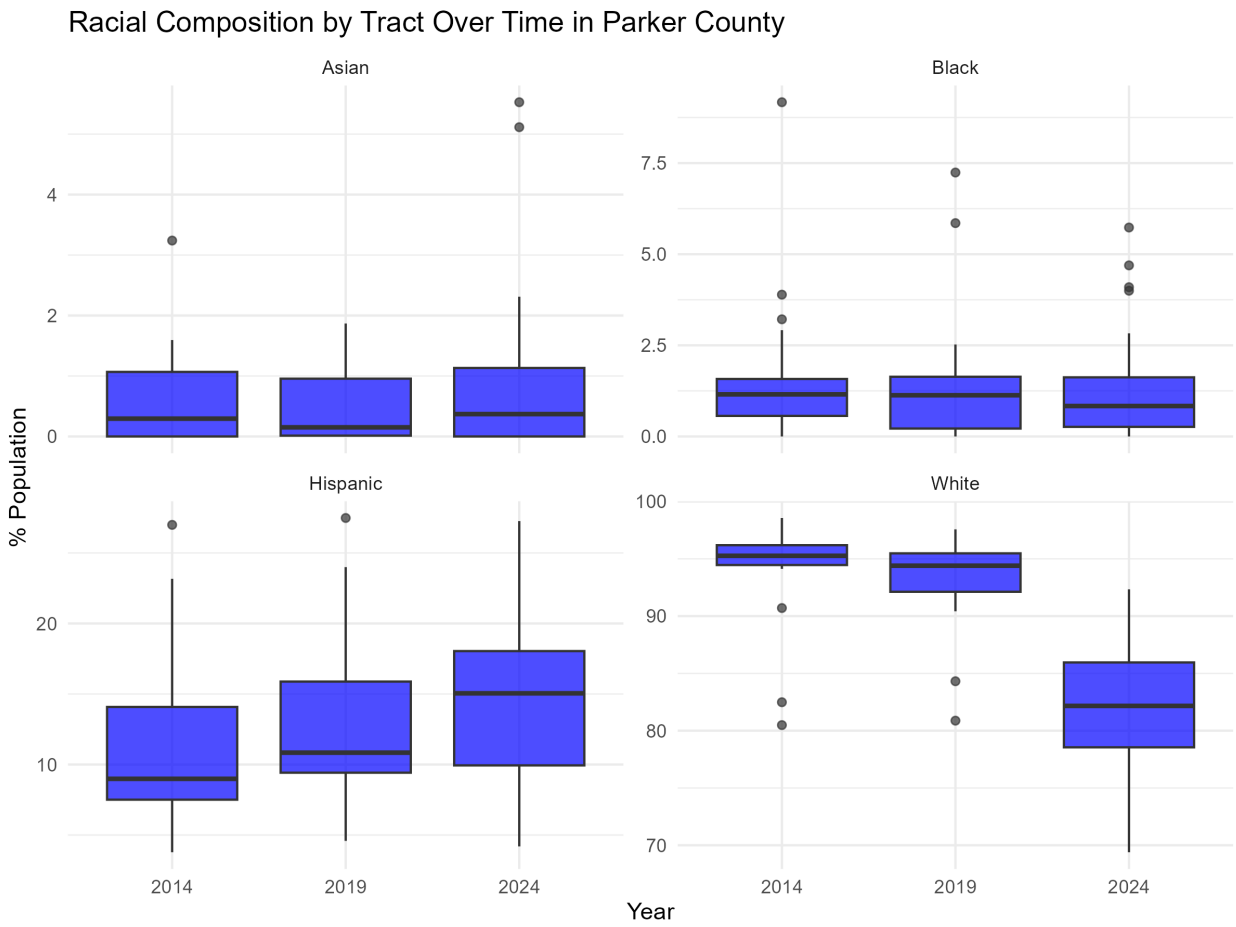
These shifts may influence:

- School demographics
- Community services
- Local government priorities
- Long-term political dynamics

Change in Hispanic Population Share (2014-2024)



Note: Gray areas indicate missing or suppressed data, often due to census tract boundary changes between 2014 and 2024 or insufficient data for reliable estimates.
Source: American Community Survey (ACS 5-Year Estimates)

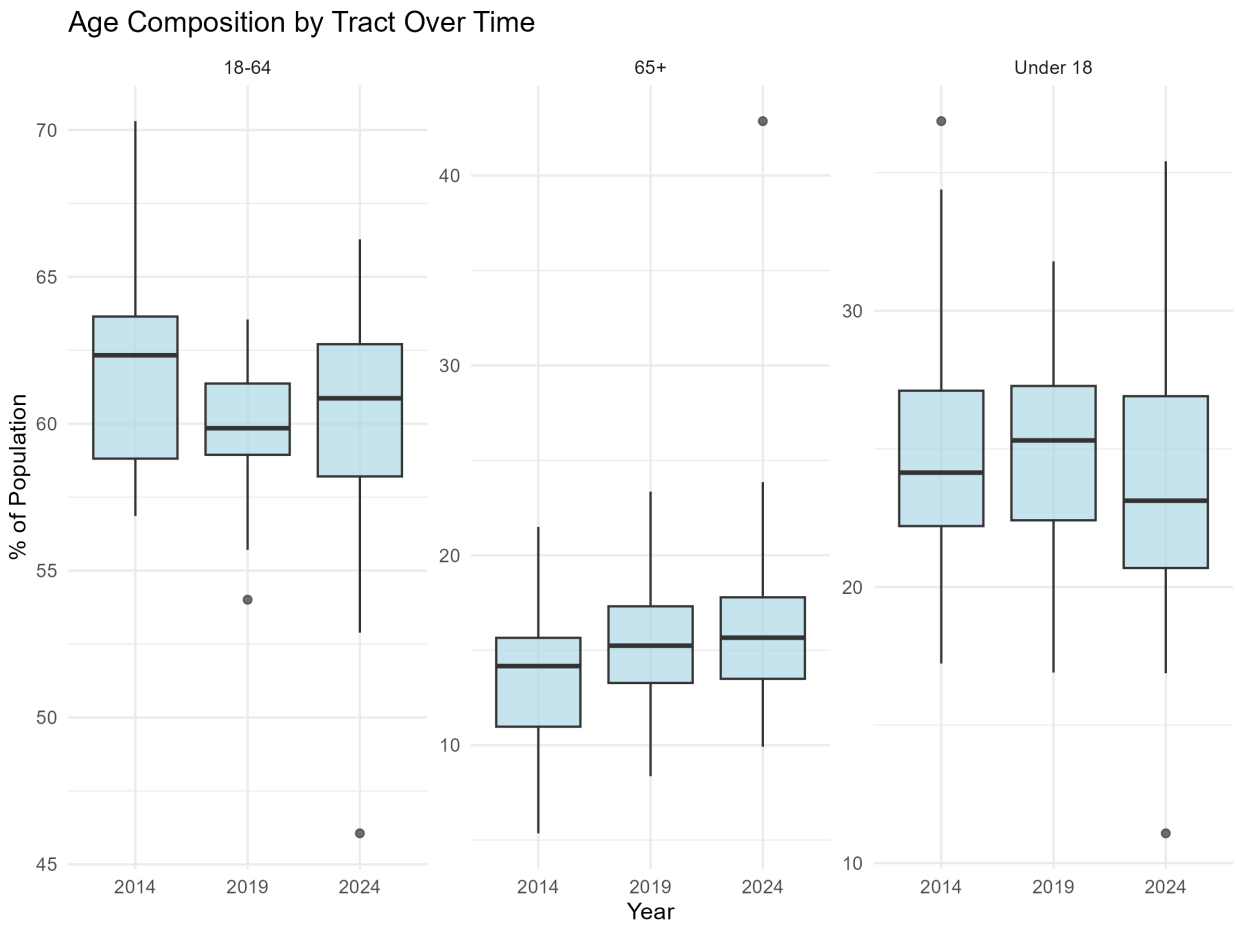


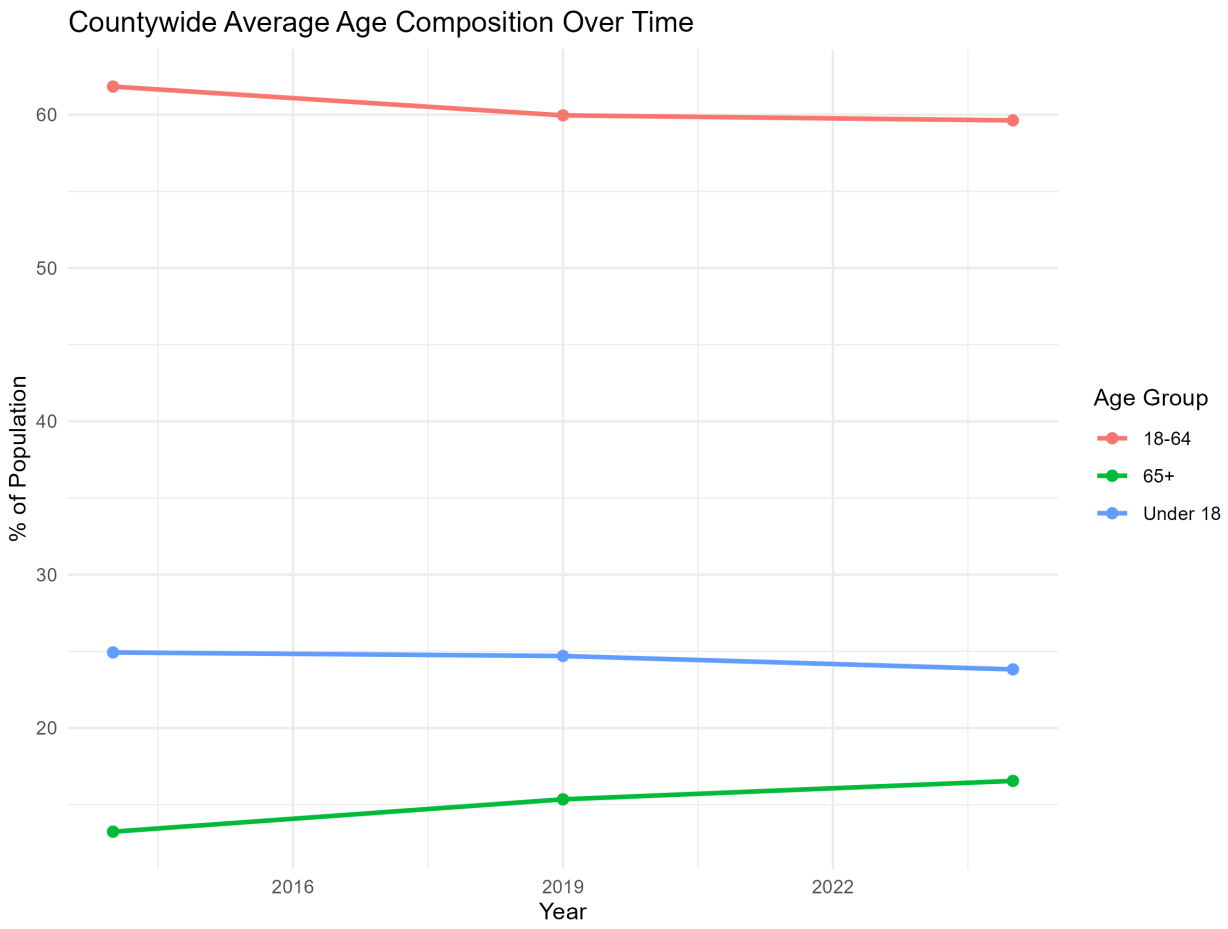
Age Distribution Patterns

The share of residents aged 65 and older increased steadily between 2014 and 2024, while the proportion under 18 and ages 18-64 declined slightly. This pattern is consistent across both tract-level distributions and countywide averages.

These results suggest:

- The population is gradually aging.
- Demand for healthcare and senior services will likely increase.
- Housing and infrastructure may need to adapt.
- Slower youth growth may impact future school enrollment.


Age Composition by Tract Over Time
18-64
65+
Under 18
% of Population
Year
70
65
60
55
50
45
40
30
20
10
30
20
10
2014
2019
2024



5. Overall Findings

Taken together, the results reveal a county undergoing accelerated suburban transformation.

Between 2014 and 2024, Parker County experienced:

- Rapid population expansion
- Significant income growth
- Increasing spatial inequality
- Rising demographic diversity
- Gradual population aging

These trends indicate structural change rather than isolated shifts. Parker County is transitioning from a primarily rural county into a rapidly growing outer-ring suburban community shaped by metropolitan expansion.

These patterns are supported by tract-level trend summaries (see Appendix).

6. Limitations

- Census tract boundaries change over time; this analysis uses 2024 boundaries for consistency.
- ACS estimates include sampling error not explicitly modeled.
- Some tracts contain missing or suppressed data.

7. Conclusion

Parker County is undergoing significant demographic and economic transformation. Growth is uneven, inequality is increasing, and the population is becoming more diverse and older. These findings highlight how suburban expansion is reshaping communities at the local level and provide a foundation for understanding future challenges related to housing, infrastructure, and public services.

Appendix

A1. Tract-Level Trend Summary

The following table reports tract-level averages for key variables across 2014, 2019, and 2024.

```r
trend_scan <- read.csv("../outputs/trend_scan.csv")

kableExtra::landscape(
  knitr::kable(trend_scan, format = "latex", booktabs = TRUE) %>%
    kable_styling(latex_options = "scale_down", font_size = 9)
)
```

year	total_pop_avg	med_income_avg	pct_under_18_avg	pct_18_64_avg	pct_65_plus_avg	pct_white_avg	pct_black_avg	pct_asian_avg	pct_hispanic_avg	pct_enrolled_avg
2014	6304.421	65821.16	24.92874	61.83272	13.23854	93.90777	1.656952	0.6663470	11.41774	73.06894
2019	7042.684	78350.00	24.69691	59.95456	15.34853	93.10607	1.524217	0.5098804	12.85222	72.93153
2024	5695.448	108759.55	23.82364	59.62614	16.55022	82.68048	1.356879	0.8988336	14.71623	72.07151

Disclaimer

This product uses the Census Bureau Data API but is not endorsed or certified by the Census Bureau.

References:

- https://www.census.gov/programs-surveys/acs/data/data-via-api.html
- https://www.parkercountyedc.com/news
- https://www.census.gov/data/developers/data-sets/acs-5year.html
- https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html
- https://fortworthreport.org/2024/08/18/fast-growing-parker-county-prepares-for-a-lot-more-change-with-arrival-of-uta-campus/#
- https://www.mckinsey.com/institute-for-economic-mobility/our-insights/small-towns-massive-opportunity-unlocking-rural-americas-potential
- https://trerc.tamu.edu/article/can-texas-sustain-its-growth/#
- https://www.fox26houston.com/news/texas-immigration-report-2025